

Possible Robotics Applications of PLUG and ILRM

Preliminary synthetic screening against GRU and candidate implementation areas

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Overview

PLUG and ILRM are experimental recurrent architectures designed to give a recurrent network a direct, fixed, inverse-lag view of earlier inputs. This note collates a set of controlled synthetic robotics-style screening experiments comparing standard GRU, PLUG, and ILRM. The purpose is not to claim that PLUG or ILRM solve robotics problems, but to identify application types in which their long-memory behaviour appears promising enough to justify testing on established robotics simulators and real datasets.

The strongest recurring pattern is narrow but potentially useful: PLUG and ILRM often performed well when a brief observation remained relevant after hundreds or thousands of later sensor updates. Examples included occluded navigation cues, payload or tool identity, delayed fault precursors, and localization during long absolute-position dropouts. In contrast, ordinary GRU was competitive or better on continuously evolving dynamics such as soft-arm hysteresis, and all three architectures struggled when the required latent physical state kept drifting far beyond the training horizon.

Experimental status. All results below are synthetic screening experiments produced during exploratory development. They are not peer-reviewed robotics benchmarks, not real-robot results, and not evidence of production safety, reliability, or general superiority over GRU, LSTM, Transformers, state-space models, or other recurrent approaches. Small seed counts were used to keep the screens bounded. Long endpoints, particularly 4,096–8,192 steps, should be treated as exploratory.

This note follows the original PLUG and ILRM publications dated 31 August 2026 and the Raw State PLUG & ILRM follow-up dated 2 September 2026. The Raw State report uses the same cautious screening convention and explicitly distinguishes synthetic persistence evidence from application-level accuracy.

1 Motivation and scope

GRUs remain useful in robotics because they are compact recurrent models that can run on streaming observations. A possible weakness arises when one brief observation remains important for a very long time: a landmark leaves the field of view, an absolute localization fix disappears, a payload remains attached for thousands of control cycles, an early vibration precursor matters much later, or an operator’s intention must persist through noisy interaction.

The intended research question here is therefore narrow:

Can a direct inverse-lag history improve GRU-like recurrent models on robotics tasks where sparse old observations remain causally relevant, without creating unacceptable penalties on ordinary or continuously evolving dynamics?

The screens below were designed to probe that boundary rather than to demonstrate a complete robot controller.

2 Architectures compared

The standard comparison used three models with approximately matched parameter counts.

2.1 GRU baseline

The GRU is the ordinary recurrent baseline. It must preserve all useful long-range information through its learned hidden state.

2.2 PLUG

PLUG supplies a standard GRU with the current input and a normalised inverse-lag history of previous inputs. At timestep t ,

$$P_t = \frac{1}{H_t} \sum_{k=1}^t \frac{x_{t-k}}{k}, \quad H_t = \sum_{k=1}^t \frac{1}{k}, \quad P_0 = 0,$$

and the GRU receives

$$e_t = [x_t; P_t].$$

The GRU recurrence itself remains unchanged. This makes PLUG the simpler candidate for adding long-history information to an existing GRU-based robotics pipeline.

2.3 ILRM

ILRM uses the same P_t but builds the recurrent update around it. Form

$$z_t = [x_t; h_{t-1}; P_t],$$

compute access and write gates

$$r_t = \sigma(W_r z_t + b_r), \quad w_t = \sigma(W_w z_t + b_w),$$

construct

$$\tilde{h}_t = \tanh(W_c[x_t; r_t \odot h_{t-1}; P_t] + b_c),$$

and update

$$h_t = (1 - w_t) \odot h_{t-1} + w_t \odot \tilde{h}_t.$$

The practical hypothesis is that PLUG may be attractive when an old observation should simply remain available, while ILRM may be more useful when persistent context must also be selectively revised.

3 Experimental protocol

The robotics screens used structured numeric feature channels rather than rendered camera images or a physics-engine robot. The initial classification batch used approximately parameter-matched models around 2.8k parameters, four-way targets with 25% chance, short training trajectories sampled from 32–64 steps, and bounded training budgets. Two paired seeds were used for the main 1,024-step comparisons. Selected long-distance checks extended to 2,048, 4,096, and 8,192 steps with one or two seeds.

A second batch tested broader continuous or changing-state problems with models around 2.9–3.1k parameters. Soft-arm hysteresis, localization dropout, and actuator degradation used two paired seeds; the human-intent screen used three seeds because it showed a particularly strong signal.

Different batches were deliberately small and were not one unified benchmark suite. Absolute values should therefore be interpreted within each task, not as cross-task scores.

4 Persistent-event robotics screens

4.1 Occluded navigation

A robot observed one of four route or landmark cues and then lost sight of it. Hundreds of changing navigation and obstacle observations followed before the robot had to choose the correct route. This probes partial observability: an old visual or semantic cue remains relevant even though the immediate sensor stream has changed substantially.

Model	64	256	1,024
GRU	25.4%	25.8%	29.7%
PLUG	100.0%	100.0%	100.0%
ILRM	100.0%	100.0%	99.8%

The synthetic result is consistent with a possible application in navigation under occlusion, where a person, landmark, doorway state, route instruction, or object location is observed once and later disappears from the field of view.

4.2 Persistent payload or tool identity

The robot received a brief early signal identifying one of four payload classes, followed by changing joint and force observations. The identity was not repeatedly revealed.

Model	64	256	1,024
GRU	25.2%	25.0%	27.9%
PLUG	100.0%	100.0%	100.0%
ILRM	100.0%	100.0%	87.3%

This suggests possible use for long-lived operating context such as payload class, tool identity, mode, calibration state, or terrain/context flags. The ILRM variation also warns that optimization reliability remains important.

4.3 Delayed fault precursor

A brief abnormal vibration or force signature occurred on one of four channels and then disappeared. The system returned to apparently normal sensor activity for hundreds of steps before the model had to identify which precursor had occurred.

Model	64	256	1,024
GRU	26.0%	23.8%	27.3%
PLUG	100.0%	99.8%	100.0%
ILRM	100.0%	99.6%	100.0%

This was one of the strongest application-style signals in the screening and motivates future testing in predictive-maintenance or delayed-fault datasets. It does not establish that real mechanical precursors will have the same structure.

4.4 Payload replacement

The robot began with payload A , later changed to payload B , and had to report the current payload. This deliberately tests whether long persistence becomes a liability when old information is obsolete.

Model	64	256	1,024
GRU	100.0%	86.5%	78.3%
PLUG	100.0%	99.8%	67.8%
ILRM	100.0%	99.8%	87.9%

The result is mixed rather than universally favourable. At 1,024 steps ILRM had the best mean, but the task showed substantial seed variation. PLUG can retain stale information too strongly, while a GRU can be very competitive when the relevant state is refreshed later in the trajectory.

5 Long-distance exploratory extension

Selected models were evaluated farther without retraining. These endpoints used only one or two seeds and are therefore exploratory rather than publication-grade estimates.

Task	Steps	GRU	PLUG	ILRM
Occluded navigation	4,096	17.2–28.1%	100.0%	81.2–100.0%
Occluded navigation	8,192	28.1–31.3%	65.6–96.9%	81.2–100.0%
Delayed fault precursor	4,096	18.8–29.7%	98.4–100.0%	95.3–100.0%
Delayed fault precursor	8,192	25.0–28.1%	65.6–68.8%	81.2–81.3%
Payload identity*	4,096	28.1%	100.0%	70.3%
Payload identity*	8,192	28.1%	62.5%	68.8%
Payload replacement*	4,096	29.7%	70.3%	70.3%

*Single long-distance seed in this exploratory extension.

The long tests suggest a practical rather than unlimited memory range. PLUG was especially clean through roughly 4,096 steps on several persistent-event tasks, while ILRM sometimes retained useful information farther. Both degraded on some 8,192-step probes. This is consistent with finite-capacity and interference limits rather than unlimited memory.

6 Broader robotics-style screens

6.1 Localization through long absolute-position dropout

A robot received an early absolute-position fix and then had to continue using noisy, biased odometry after the absolute source disappeared. Lower RMSE is better.

Model	64 RMSE	256 RMSE	1,024 RMSE
GRU	0.223	0.403	0.432
PLUG	0.228	0.340	0.366
ILRM	0.231	0.339	0.361

At 1,024 steps ILRM reduced RMSE by roughly 16% relative to the GRU in this screen. This motivates testing on real or established simulated localization problems involving GNSS, UWB, visual, or beacon dropout while odometry/IMU observations continue.

6.2 Human intent with change of mind

A human provided an early intent cue, later moved ambiguously, and in roughly half of cases changed to a different final intention. The model had to report the final intent. This tests both persistence and revision.

Model	64	256	1,024
GRU	99.2%	81.8%	93.5%
PLUG	99.5%	73.7%	79.7%
ILRM	99.5%	86.5%	99.2%

ILRM’s three 1,024-step seeds were 100.0%, 100.0%, and 97.7%. In this particular screen, the learned ILRM gates appeared better suited than PLUG to information that should persist but can later be revised. This is promising for operator instructions, intention tracking, or human-robot interaction, but needs established HRI datasets before stronger claims are justified.

6.3 Soft-arm hysteresis

A continuously driven synthetic soft arm used pressure-command history to generate a hysteretic 2-D tip state. Lower RMSE is better. At 1,024 steps, the mean RMSE was:

$$\text{GRU} = \mathbf{0.194}, \quad \text{PLUG} = 0.239, \quad \text{ILRM} = 0.243.$$

This screen did **not** favour PLUG or ILRM. When every recent control input contributes continuously to a physical state, the ordinary recurrent hidden state may be better matched to the dynamics than a fixed long-tailed history.

6.4 Gradual actuator degradation

A synthetic actuator efficiency drifted over time while commands, tracking error, temperature, and current-like features changed. At 64 steps all models achieved RMSE around 0.013. At 256, PLUG and ILRM were slightly lower than GRU (approximately 0.117–0.119 versus 0.129). At 1,024, all three failed similarly:

$$\text{GRU} = 0.537, \quad \text{PLUG} = 0.539, \quad \text{ILRM} = 0.525.$$

This suggests that preserving old information is not the same as extrapolating a continuously changing physical law far beyond the training horizon.

7 Possible implementation areas

The synthetic results suggest that PLUG and ILRM should be investigated first in robotics systems where memory is *sparse, long-lived, and event-like*, rather than in every recurrent control problem.

Candidate field	Why long memory may matter	Screening indication
Occluded navigation / partial observability	Remember a landmark, person, object, doorway state, or route cue after it leaves view	Strong for PLUG and ILRM
Predictive maintenance / fault precursor memory	Preserve a brief abnormal sensor signature through long normal-looking operation	Strong for PLUG and ILRM
Localization during GNSS/UWB/vision dropout	Keep an old absolute fix or calibration relevant while noisy odometry continues	Moderate improvement for PLUG/ILRM
Payload, tool, or operating-mode memory	A state changes infrequently but affects many later control cycles	Strong for PLUG; ILRM more variable
Human intent / instruction tracking	Remember an early cue but allow later evidence to revise the intended action	Strongest signal for ILRM
Generic soft-body or motor dynamics	Continuous recent history determines current physical state	GRU favoured or no clear advantage
Continuously drifting degradation	Latent state changes throughout deployment and must be extrapolated	No convincing advantage

A useful working distinction from these screens is:

- **PLUG:** a simple candidate for long-lived observation or event memory in an existing GRU pipeline.
- **ILRM:** a candidate for long-lived but editable task/context memory when information may later need revision.

8 Implementation considerations

PLUG is the lower-disruption implementation. The history P_t is concatenated with the ordinary current input and supplied to a conventional GRU. Existing downstream heads and most training infrastructure can remain unchanged.

ILRM replaces the recurrent cell and uses the same history in its gates and candidate update. The extra structure may be useful when persistent information has to be revised, but it introduces a less familiar recurrent implementation and can show greater optimization variability on some screens.

For fixed training sequences, the inverse-lag history can be calculated as a causal convolution using a leading-zero kernel

$$q = [0, 1, 1/2, 1/3, \dots]$$

with sufficient zero-padding for linear FFT convolution, followed by harmonic normalisation. For live robot streaming, recomputing a full FFT every timestep would be inefficient; an online, chunked, or partitioned convolution implementation should be evaluated for real-time deployment. The screening results here do not include an embedded-hardware latency study.

The current results also suggest that event-related channels should remain clearly represented in the input vector. The architectures do not solve arbitrary source separation: several old useful signals can still interfere inside a finite-dimensional compressed history.

9 What the experiments currently support

The combined robotics screening supports the following narrow observations:

- PLUG and ILRM can preserve sparse task-relevant observations far beyond the 32–64-step training range in several controlled robotics-style tasks.

- Occluded navigation and delayed fault-precursor memory produced the clearest long-distance separation from GRU.
- PLUG was particularly reliable on persistent external cues such as route/landmark information and payload identity.
- ILRM showed a promising advantage on human-intent revision and a modest localization-dropout improvement, suggesting that its learned gates may help when long-lived context must also be edited.
- The advantage is not universal: GRU was better on the soft-arm hysteresis screen, and no architecture solved the long extrapolation of continuously drifting actuator degradation.
- The 8,192-step probes show degradation and variability, so these models should not be interpreted as unlimited-memory systems.

10 What the experiments do not support

These screens do not establish that PLUG or ILRM:

- improve real robot navigation, manipulation, localization, HRI, or predictive maintenance;
- are safer or more reliable than standard recurrent models in production;
- universally outperform GRU, LSTM, Transformers, state-space models, attention, or external-memory systems;
- solve continuously evolving physical dynamics merely because those dynamics have long histories;
- solve arbitrary multi-item binding, exact event reconstruction, or stale-memory interference;
- maintain perfect memory indefinitely.

The results should instead be treated as a prioritisation tool for selecting realistic benchmarks where the architecture’s demonstrated persistence mechanism is plausibly relevant.

11 Recommended next validation

A compact next-stage robotics programme would test three established application families:

1. **Visual or LiDAR navigation with occlusion / partial observability.** Compare success rate and collision rate as occlusion duration exceeds the training horizon.
2. **Localization with controlled sensor dropout.** Compare trajectory RMSE and drift when GNSS/UWB/vision fixes disappear for increasingly long intervals.
3. **Human-intent or instruction tracking with revisions.** Compare final-intent accuracy when early cues are later confirmed, contradicted, or replaced.

A predictive-maintenance dataset with rare early precursors is also a strong candidate if sufficiently long labelled sensor histories are available. Future comparisons should use paired seeds, matched parameter budgets, common training data, and enough repeated runs to quantify the optimization variability seen in these screens.

12 Publication, contact and licensing

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